

A FAITHFUL COTANGENT CLASS OVER A LOCAL ARTINIAN RING IN CHARACTERISTIC ZERO

ABSTRACT. Let k be a field of characteristic zero. We construct a local Artinian k -algebra R of length 176, with residue field k , and an explicit class $\xi \in \pi_1(L_{R/k})$ such that $\text{Ann}_R(\xi) = 0$. The proof reduces faithfulness to a Lefschetz isomorphism in $k[X_1, \dots, X_{10}]/(X_1^2, \dots, X_{10}^2)$.

Let

$$\begin{aligned} P &= k[x_1, \dots, x_{10}]_{(x_1, \dots, x_{10})}, & \mathfrak{n} &= (x_1, \dots, x_{10})P, \\ e_4(x_1, \dots, x_{10}) &:= \sum_{1 \leq i < j < \ell < m \leq 10} x_i x_j x_\ell x_m, & F &:= \sum_{i=1}^{10} x_i^3 + e_4(x_1, \dots, x_{10}), \\ g_i &= \frac{\partial F}{\partial x_i}, \end{aligned}$$

and set

$$I = (g_1, \dots, g_{10}) + \mathfrak{n}^4, \quad R = P/I.$$

Theorem 1. *The ring R is local Artinian with residue field k and length 176. The class*

$$\xi = [F] \in \pi_1(L_{R/k})$$

has zero annihilator.

We use the following standard fact. Let $(T, \mathfrak{t})$ be a Noetherian local ring, equipped with its $\mathfrak{t}$ -adic filtration, and suppose that the initial forms $f_1^*, \dots, f_r^*$ of $f_1, \dots, f_r \in \mathfrak{t}$ form a regular sequence in $\text{gr}_{\mathfrak{t}} T$. If $\bar{\mathfrak{t}} = \mathfrak{t}/(f_1, \dots, f_r)$, then the natural map induces an isomorphism

$$\frac{\text{gr}_{\mathfrak{t}} T}{(f_1^*, \dots, f_r^*)} \xrightarrow{\sim} \text{gr}_{\bar{\mathfrak{t}}} (T/(f_1, \dots, f_r)).$$

Sketch of proof: First take one element f , put $a = \text{ord}_{\mathfrak{t}}(f)$, and assume that f^* is a nonzerodivisor on $\text{gr}_{\mathfrak{t}} T$. Then

$$\text{ord}_{\mathfrak{t}}(fh) = a + \text{ord}_{\mathfrak{t}}(h), \quad (fh)^* = f^* h^*$$

for every nonzero $h \in T$. With the convention $\mathfrak{t}^j = T$ for $j \leq 0$, this gives

$$(f) \cap \mathfrak{t}^d = f \mathfrak{t}^{d-a}$$

for every integer d , and hence the exact sequence

$$0 \longrightarrow (\text{gr}_{\mathfrak{t}} T)(-a) \xrightarrow{f^*} \text{gr}_{\mathfrak{t}} T \longrightarrow \text{gr}_{\bar{\mathfrak{t}}} (T/(f)) \longrightarrow 0.$$

The general case follows by induction.

Proof of Theorem 1.

The associated graded ring: Equip P with its $\mathfrak{n}$ -adic filtration. Put

$$G = (g_1, \dots, g_{10}), \quad S = P/G, \quad \mathfrak{q} = \mathfrak{n}S,$$

and equip S with its $\mathfrak{q}$ -adic filtration. The initial form of g_i is $3X_i^2$ in

$$\text{gr}_{\mathfrak{n}} P \cong k[X_1, \dots, X_{10}].$$

The sequence $X_1^2, \dots, X_{10}^2$ is regular, so the standard filtered fact above gives

$$(1) \quad A := \text{gr}_{\mathfrak{q}} S \cong k[X_1, \dots, X_{10}]/(X_1^2, \dots, X_{10}^2).$$

Since $R = S/\mathfrak{q}^4$, its maximal ideal is $\mathfrak{m} = \mathfrak{q}/\mathfrak{q}^4$. We equip R with its $\mathfrak{m}$ -adic filtration. Equation (1) gives

$$(2) \quad \mathrm{gr}_{\mathfrak{m}} R \cong A/A_{\geq 4} = A_0 \oplus A_1 \oplus A_2 \oplus A_3.$$

The squarefree monomials form a basis of A , hence

$$\dim_k R = \sum_{d=0}^3 \binom{10}{d} = 176.$$

Thus R is local Artinian, $R/\mathfrak{m} = k$, and $\mathfrak{m}^4 = 0$.

The socle: Clearly $\mathfrak{m}^3 \subseteq \mathrm{Soc}(R)$. Conversely, suppose $0 \neq r \in \mathrm{Soc}(R)$ has initial form $0 \neq \rho \in A_d$ for some $d < 3$. Write its unique squarefree expansion as

$$\rho = \sum_{|T|=d} c_T X_T.$$

Choose T with $c_T \neq 0$ and $i \notin T$. The coefficient of $X_{T \cup \{i\}}$ in $X_i \rho$ is c_T , so $X_i \rho \neq 0$, contradicting $x_i r = 0$. Therefore

$$(3) \quad \mathrm{Soc}(R) = \mathfrak{m}^3 \cong A_3.$$

The cotangent class: Euler's identity gives

$$\sum_{i=1}^{10} x_i g_i = 3 \sum_{i=1}^{10} x_i^3 + 4e_4(x_1, \dots, x_{10}), \quad 3F = \sum_{i=1}^{10} x_i g_i - e_4(x_1, \dots, x_{10}),$$

so $F \in I$. Since P is smooth over k , transitivity gives an exact sequence

$$0 \longrightarrow \pi_1(L_{R/k}) \longrightarrow I/I^2 \xrightarrow{d} \Omega_{P/k} \otimes_P R.$$

Now $dF = \sum_i g_i dx_i$, which vanishes in the target. Hence F defines a class

$$\xi = [F] \in \pi_1(L_{R/k}).$$

The socle acts faithfully on ξ : The map $P \rightarrow S$ induces an R -linear map

$$(4) \quad \Psi : I/I^2 \longrightarrow \mathfrak{q}^4/\mathfrak{q}^8.$$

In S one has $F = -\frac{1}{3}e_4(x_1, \dots, x_{10})$, and hence

$$\Psi(s\xi) = -\frac{1}{3}s e_4(x_1, \dots, x_{10}) \in \mathfrak{q}^7/\mathfrak{q}^8 \quad (s \in \mathrm{Soc}(R)).$$

Thus, via (3), it suffices to prove that

$$(5) \quad \cdot e_4 : A_3 \longrightarrow A_7$$

is an isomorphism. Let $\ell = X_1 + \dots + X_{10}$ and define

$$\Lambda(X_T) = \sum_{i \in T} X_{T \setminus \{i\}}.$$

A direct calculation gives

$$[\Lambda, \ell]|_{A_d} = (10 - 2d) \mathrm{id}_{A_d}.$$

Thus ℓ , Λ , and the grading operator give the standard $\mathfrak{sl}_2$ -action on A . After extending scalars to an algebraic closure, decomposition into irreducible $\mathfrak{sl}_2$ -modules gives

$$\ell^4 : A_3 \xrightarrow{\sim} A_7.$$

Since $X_i^2 = 0$ in A , one has $\ell^4 = 4!e_4$; this proves (5) in characteristic zero.

It follows that $\Psi(s\xi) \neq 0$ and hence $s\xi \neq 0$ for $0 \neq s \in \mathrm{Soc}(R)$.

Faithfulness: Suppose that $J = \text{Ann}_R(\xi)$ is nonzero. Choose t maximal such that $\mathfrak{m}^t J \neq 0$. Then

$$0 \neq \mathfrak{m}^t J \subseteq \text{Soc}(R),$$

and every element of $\mathfrak{m}^t J$ annihilates ξ , a contradiction. Hence $\text{Ann}_R(\xi) = 0$. $\square$